# Lecture 1 Exercise Solutions

1. Prove directly, from the very definition of divisibility, that if $d|x$ and $d|y$, then $d|x-y$ and $d|x+y$.

   Proof: As $d$ divides $x$, $x=k_1 d$, and as $d$ divides $y$, $y=k_2 d$, then $x\pm y=k_1 d\pm k_2 d=(k_1\pm k_2)d$, thus, $d$ divides $x\pm y$.

2. Show that for every $n$, $n+1$ and $n$ are relatively prime.

   If $n$ is an odd number, then $n$ and $n+1$ are obviously relatively prime, since 2 only divides $n+1$ and any subsequent number that divides $n$ would also have to divide 1 and that is impossible.

   If $n$ is even, then $n=kd$ and, thus, $n+1=kd+1$. So $n+1$ is not divisible by $d$ and therefore $n$ and $n+1$ are relatively prime.

3. Show that for every $n$ that $n$ and $n^2+1$ are relatively prime.

   Let $d$ be a divisor of $n$, so $n=kd \Rightarrow n^2+1=k^2d^2+1$ and this number is not divisible by $d$.

4. Factor $1028$ into primes.

   $2\cdot 2\cdot 257=1028$

5. Factor $2057$ into primes.

   $11\cdot 11\cdot 17=2057$

6. Find a proper factor of $111,111,111,111,111$.

   $111,111,111,111,111=111\cdot 10^{12}+111\cdot 10^9+111\cdot 10^6+111\cdot 10^3+111=$
   $111\cdot(10^{12}+10^9+10^6+10^3+1)$

   So, $111$ is a proper factor.

7. Find a proper factor of $110,111,101,111,011$.

   $110,111,101,111,011=11011\cdot 10^{10}+11011\cdot 10^5+11011=11011\cdot(10^{10}+10^5+1)$

So, $11011$ is a proper factor.

8. Find a proper factor of $101{,}010{,}101{,}010{,}101$.

   $101{,}010{,}101{,}010{,}101 = 1010101 \cdot 10^8 + 1010101 = 1010101 \cdot (10^8 + 1)$

   So, $1010101$ is a proper factor.

9. Explain why $n^2 - 1$ cannot be prime for any $n > 2$.

   $n^2 - 1 = (n-1)(n+1)$

10. Explain why $3^n - 1$ cannot possibly be a prime number if $n > 1$.

    $3^n - 1 = (3-1) \cdot (3^{n-1} + \ldots + 1)$

11. Find $\gcd(102313, 103927)$ without factoring.

    Using Euclidean Algorithm we obtain

    $103927 - 1 \cdot 102313 = 1614$
    $102313 - 63 \cdot 1614 = 631$
    $1614 - 2 \cdot 631 = 352$
    $631 - 1 \cdot 352 = 279$
    $352 - 1 \cdot 279 = 73$
    $279 - 3 \cdot 73 = 60$
    $73 - 1 \cdot 60 = 13$
    $60 - 4 \cdot 13 = 8$
    $13 - 1 \cdot 8 = 5$
    $8 - 1 \cdot 5 = 3$
    $5 - 1 \cdot 3 = 2$
    $3 - 1 \cdot 2 = 1$

    So, $\gcd(102313, 103927) = 1$

12. Based on the previous problem find the inverse of $\overline{102{,}313}$ in $Z^{\times}_{103927}$

    $102313$ and $103927$ are relatively prime. Thus, there is the inverse of $\overline{102{,}313}$ in $Z^{\times}_{103927}$. To find the inverse of $\overline{102{,}313}$ we have to present $102313$ and $103927$ in the form of $a \cdot 102313 + b \cdot 103927 = 1$, which is possible because of

$\gcd(102313,103927) = 1$. Therefore, $a$ will be the inverse. Indeed, $a \cdot 102313 = (-b) \cdot 103927 + 1$, i.e. $(a \cdot 102313)\%103927 = 1$

Using calculations of the Euclidean Algorithm in the opposite order and substituting, we obtain

$1 = 1 \cdot 3 - 1 \cdot 2 = 1 \cdot 3 - 1 \cdot (5 - 1 \cdot 3) = -1 \cdot 5 + 2 \cdot 3$

$1 = -1 \cdot 5 + 2 \cdot (8 - 1 \cdot 5) = 2 \cdot 8 + (-3) \cdot 5$

$1 = 2 \cdot 8 + (-3) \cdot 5 = 2 \cdot 8 + (-3) \cdot (13 - 1 \cdot 8) = -3 \cdot 13 + 5 \cdot 8$

$1 = -3 \cdot 13 + 5 \cdot 8 = -3 \cdot 13 + 5 \cdot 8 = -3 \cdot 13 + 5 \cdot (60 - 4 \cdot 13) = 5 \cdot 60 + (-23) \cdot 13)$

$1 = 5 \cdot 60 + (-23) \cdot 13 = 5 \cdot 60 + (-23) \cdot (73 - 1 \cdot 60) = -23 \cdot 73 + 28 \cdot 60$

$1 = -23 \cdot 73 + 28 \cdot 60 = -23 \cdot 73 + 28 \cdot (279 - 3 \cdot 73) = 28 \cdot 279 + (-107) \cdot 73$

$1 = 28 \cdot 279 + (-107) \cdot 73 = 28 \cdot 279 + (-107) \cdot (352 - 1 \cdot 279) = (-107) \cdot 352 + 135 \cdot 279$

$1 = (-107) \cdot 352 + 135 \cdot 279 = (-107) \cdot 352 + 135 \cdot (631 - 1 \cdot 352) = 135 \cdot 631 + (-242) \cdot 352$

$1 = 135 \cdot 631 + (-242) \cdot 352 = 135 \cdot 631 + (-242) \cdot (1614 - 2 \cdot 631) = -242 \cdot 1614 + 619 \cdot 631$

$1 = -242 \cdot 1614 + 619 \cdot 631 = -242 \cdot 1614 + 619 \cdot (102313 - 63 \cdot 1614) = 619 \cdot 102313 + (-39239) \cdot 103927$

$1 = 619 \cdot 102313 + (-39239) \cdot 1614 = 619 \cdot 102313 + (-39239) \cdot (103297 - 1 \cdot 102313) = -39239 \cdot 103927 + 39858 \cdot 102313$

So, $a \cdot 102313 + b \cdot 103927 = 39858 \cdot 102313 + (-39239) \cdot 103927$, therefore $a = 39858$, meaning $\overline{39858}$ is the inverse of $\overline{102313}$ in $Z_{103927}^{\times}$.

13. For an integer n, show that the greatest common divisor of the integers $n^3 + n^2 + n + 1$ and $n^2 + n + 1$ is just 1.

$n^3 + n^2 + n + 1 = n^2(n+1) + (n+1) = (n^2+1)(n+1)$, so that $n^2 + 1$ and $n + 1$ are the only divisors of $n^3 + n^2 + n + 1$. But $n^2 + n + 1$ has no divisors at all. Thus the $\gcd(n^3 + n^2 + n + 1, n^2 + n + 1) = 1$.

14. For an integer $n > 0$, what is greatest common divisor of $n^3 + n^2 + n + 1$ and $n^8 + n^7 + n^6 + n^5 + n^4 + n^3 + n^2 + n + 1$?

$n^3 + n^2 + n + 1 = (n^2 + 1)(n + 1)$

$n^8+n^7+n^6+n^5+n^4+n^3+n^2+n+1=$
$(n^8+n^7)+(n^6+n^5)+(n^4+n^3)+(n^2+n)+1=$
$n^7(n+1)+n^5(n+1)+n^3(n+1)+n(n+1)+1=$
$(n^7+n^5+n^3+n)(n+1)+1=$
$n(n^6+n^4+n^2+1)(n+1)+1=$
$n[n^4(n^2+1)+(n^2+1)](n+1)+1=$
$n(n^2+1)(n^4+1)(n+1)+1=$
$n(n^4+1)(n^2+1)(n+1)+1$

Both numbers are divisible by $n+1$ and $n^2+1$ (i.e., they are common divisors). But, $\forall n \in Z(n^2+1>n+1)$ for $n>0$. Thus,
$\gcd(n^8+n^7+n^6+n^5+n^4+n^3+n^2+n+1, n^3+n^2+n+1)=1$.

15. By trial and error, find a multiplicative inverse to $3 \bmod 7$.

$3\cdot 5=2\cdot 7+1 \Rightarrow 5$ is the inverse.

16. Find the multiplicative inverse to $5 \bmod 7$.

We can solve his using the Extended Euclidean algorithm. First start with the Euclidean algorithm:

gcd(7,5): $7=1\cdot 5+2$
gce(5,3): $5=2\cdot 2+1$

Now let's reverse it with the Extended Euclidean:

Start: $1=5-2\cdot 2$
Substitute: $1=5-2\cdot(7-1\cdot 5)$
Simplify: $1=3\cdot 5+(-2)\cdot 7$

Clearly $3\cdot 5+(-2)\cdot 7=1$ is our solution. What does this mean? Well, let's put in a form that we are accustomed to seeing, which is the very definition of modular arithmetic:

$x=q\cdot m+r$

The above is nothing more than

$3\cdot 5=2\cdot 7+1$

So, $x = 3 \cdot 5$, $q = 2$, $m = 7$, and $r = 1$. But this is just a long way of expressing $x\%m = r$.

So, $(3 \cdot 5)\%7 = 1$. Obviously, this is the very definition of a multiplicative inverse these two numbers when multiplied by each other is 1 (mod 7). So the multiplicative inverse of 5 is 3 (mod 7).

There's another, quicker way of calculating an inverse if we know the order of the group. We will learn that $x^{\varphi(n)} = 1(\mathrm{mod}\, n)$. $\varphi(7) = 6$, so

$$x^{\varphi(n)} = 1(\mathrm{mod}\, n)$$
$$5^6 = 1(\mathrm{mod}\, n)$$
$$5 \cdot 5^{6-1} = 1(\mathrm{mod}\, n)$$

So the inverse of $5$ is $5^{6-1} = 5^5(\mathrm{mod}\, 7)$

$$5^5\%7 = 3125\%7 = 3\%7$$

So, again, 3 is the inverse of 5 (mod 7). Same result.

Obviously, just this simple small group led to a relatively large number $5^5 = 3125$. Imagine what this would look like for numbers on the order of 1000 bits or more. Fortunately, we have a solution to this problem that you'll learn about later.

17. Find the multiplicative inverse to $7 \bmod 11$.

    $7 \cdot 8 = 5 \cdot 11 + 1 \Rightarrow 8$ is the inverse.

18. Find the multiplicative inverse to $5 \bmod 13$.

    $5 \cdot 8 = 3 \cdot 13 + 1 \Rightarrow 8$ is the inverse.

19. Find the multiplicative inverse to $9 \bmod 11$.

    $9 \cdot 5 = 4 \cdot 11 + 1 \Rightarrow 5$ is the inverse.

20. Find the multiplicative inverse to $n(\mathrm{mod}\, 2n-1)$.

    $n$ and $2n-1$ must be relatively prime (and they are), so there exist $a$ and $b$, such that $an + b(2n-1) = 1$.

    Let's find $a$ and $b$:

$an + b(2n-1) = 1 \Rightarrow an + 2bn - b = 1 \Rightarrow (a+2b)n - b = 1 \Rightarrow$
$a + 2b = 0$ and $-b = 1 \Rightarrow b = -1$ and , thus, $a - 2 = 0 \Rightarrow a = 2$

Substituting into $an + b(2n-1) = 1,$ we get $2n - (2n-1) = 1$ and, therefore, $2$ is the multiplicative inverse to $n \bmod 2n-1.$ Indeed, $2n + (-1)\cdot(2n-1) = 1.$

21. Find the multiplicative inverse to $n \bmod 2n+1.$

Let's find $a$ and $b$, such that $an + b(2n+1) = 1 \Rightarrow an + 2bn + b = 1 \Rightarrow$
$(a+2b)n + b = 1 \Rightarrow b = 1$ and $a + 2b = 0 \Rightarrow a = -2.$ So, substituting,
$-2n + 1(2n+1) = 1 \Rightarrow -2$ is the multiplicative inverse to $n \bmod 2n+1.$

22. Find the multiplicative inverse to $n \bmod 3n+1.$

$an + b(3n+1) = 1 \Rightarrow an + 3bn + b = 1 \Rightarrow (a+3b)n + b = 1 \Rightarrow b = 1$ and
$a + 3b = 0 \Rightarrow a + 3 = 0 \Rightarrow a = -3.$ Thus, $-3n + 1\cdot(3n+1) = 1,$ so $-3$ is the inverse to $n \bmod 3n+1.$

23. Find the multiplicative inverse to $n \bmod n^2+1.$

$an + b(n^2+1) = 1 \Rightarrow an + bn^2 + b = 1 \Rightarrow (a+bn)n + b = 1 \Rightarrow b = 1$ and
$a + bn = 0 \Rightarrow a + n = 0 \Rightarrow a = -n.$ Thus, $(-n)n + 1(n^2+1) = 1,$ so $-n$ is the inverse to $n \bmod n^2+1.$

24. Find the multiplicative inverse to $n \bmod n^2-1.$

$an + b(n^2-1) = 1 \Rightarrow an + bn^2 - b = 1 \Rightarrow (a+bn)n - b = 1 \Rightarrow b = -1$ and
$a + bn = 0 \Rightarrow a = n,$ thus, $n\cdot n + (-1)(n^2-1) = 1.$ So, $n$ is the multiplicative inverse to $n \bmod n^2-1.$

25. Find the multiplicative inverse of 21 in $Z_{25}^{\times}.$

$$25 = 1\cdot 21 + 4 \Rightarrow \begin{pmatrix} 21 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 25 \\ 21 \end{pmatrix}$$

$$21 = 5\cdot 4 + 1 \Rightarrow \begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -5 \end{pmatrix}\begin{pmatrix} 21 \\ 4 \end{pmatrix}$$

$$4 = 4\cdot 1 + 0 \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Thus,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & -5 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 25 \\ 21 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 & 6 \\ 1 & -4 \end{pmatrix}\begin{pmatrix} 25 \\ 21 \end{pmatrix}$$

and $1 = -5 \cdot 25 + 6 \cdot 21$. Therefore, $6$ is the inverse of $21$ in $Z_{25}^{\times}$.

26. Find an integer $x$ so that $(x\%1009)\%1013 \neq x\%1013$.

$x = 1014$

27. Take two positive integers $n$ and $N$ with $n$ not dividing $N$. Find an integer $x$ so that $(x\%N)\%n \neq x\%n$.

Let $x = \max\{n, N\} + 1$. Then for $N > n$ or $n > N$ $(x\%N)\%n \neq x\%n$.

28. How many elements does the set $Z_n$ have?

$$Z_n^{\times} = \left\{ \overline{0}, \quad \overline{1}, \quad \overline{2}, \quad \ldots, \quad \overline{n-1} \right\}$$

So, $Z_n$ has $n$ elements.

29. How many elements does the set $Z_{30}^{\times}$ have?

$$Z_{30}^{\times} = \left\{ \overline{1}, \quad \overline{7} \quad \overline{11}, \quad \overline{13}, \quad \overline{17}, \quad \overline{19}, \quad \overline{23}, \quad \overline{29} \right\}$$

So we have $\left|Z_{30}^{\times}\right| = 8$ elements in this set. The same result could be obtained by the following way:

If $n = p_1^{e_1} \ldots p_n^{e_n}$ then $\varphi(n) = (p_1 - 1)p_1^{e_1 - 1} \ldots (p_n - 1)p_n^{e_n - 1}$. So, for $n = 30$
$30 = 2^1 \cdot 3^1 \cdot 5^1 \Rightarrow \varphi(30) = 1 \cdot 2 \cdot 4 = 8$.

30. Compute $2^{1000}\%11$.

We know that $(xy)\%m = ((x\%m)(y\%m))\%m$, so generalizing it we obtain

$(x_1 \cdot x_2 \cdot \ldots \cdot x_m)\%m = ((x_1\%m)\cdot(x_2\%m)\cdot\ldots\cdot(x_n\%m))\%m$

Applying this

$2^{1000}\%11 = (2^{10}\cdot\ldots\cdot 2^{10})\%11 = ((2^{10}\%11)\cdot(2^{10}\%11)\cdot\ldots\cdot(2^{10}\%11))\%11 = 1$

Since $2^{10}\%11 = 1$

$2^{1000}\%11 = ((1\%11)(1\%11)\ldots(1\%11))\%11 = 1$

31. Compute $3^{1000}\%11$.

$3^{1000}\%11 = (3^{10}\cdot\ldots\cdot 3^{10})\%11 = ((3^{10}\%11)\cdot(3^{10}\%11)\cdot\ldots\cdot(3^{10}\%11))\%11 = 1$.

Since $3^{10}\%11 = 1$, $3^{1000}\%11 = 1$.

32. Find three distinct classes $x \bmod 105$, so that $x^2 \equiv 1 \bmod 105$.

$x^2 \equiv 1 \bmod 105 \Rightarrow x^2 = 105n + 1$

Since $105n$ ends either in $0$ or $5$, that means $105n+1$ will end in $1$ or $6$. So, just examine the squares of positive integers less than 105 and see if they end in 1 or 6. Here's one:

$41^2 = 105\cdot 16 + 1$

So, $\overline{41}$ is a root. But, if $\overline{41}$ is a root, then $\overline{-41}$ is another root and $\overline{-41} = \overline{64}$. So $\overline{41}$ and $\overline{64}$ are two roots. We have the first and second roots.

The third and the fourth roots are obviously $\bar{1}$ and $\overline{-1}$, but $\overline{-1} = \overline{104}$, so $\bar{1}$ and $\overline{104}$ are also roots.

33. Find three distinct classes $x \bmod 143$, so that $x^2 \equiv 1 \bmod 143$.

$x^2 \equiv 1 \bmod 143 \Rightarrow x^2 = 143n + 1$, so we have to find $n$

The first root is obvious, as for $n = 1$, $x^2 = 144$ and $x = 12$. So the first root is $\overline{12}$ and the second is $\overline{-12}$, but $\overline{-12} = \overline{131}$, so the first two roots are $\overline{12}$ and $\overline{131}$.

The third and the fourth roots are obviously $\bar{1}$ and $\overline{-1}$, but $\overline{-1} = \overline{142}$, so that $\bar{1}$ and $\overline{142}$ are also roots.